



Yoonhyuk Choi (Last update: Sep. '26)

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RESEARCH INTEREST

My research interests lie in **Machine Learning** and **Data Mining**, particularly in *Large Language Models*, *Graph Neural Networks*, and *Recommender Systems*.

EDUCATION

- **Seoul National University** Mar. 2019 - Aug. 2023
Ph.D. & M.S., Computer Engineering
Advisor: [Chong-Kwon Kim](#)
- **University of Seoul** Mar. 2013 - Feb. 2019
B.S., Computer Science
Advisor: [Eui-Kyeong Hong](#)

WORK EXPERIENCE

- **Assistant Professor** Sep. 2025 -
Sookmyung Women's University, Seoul, South Korea
Department of AI
- **Research scientist** Oct. 2024 - Aug. 2025
Samsung SDS, Seoul, South Korea
Preceding AI Lab (LLM, RAG)
- **Postdoc. associate** Nov. 2023 - Sep. 2024
Arizona State University, Tempe, United States (PI: [Selcuk Candan](#) and [Huan Liu](#))
School of Computing and Augmented Intelligence (SCAI)
- **Postdoc. associate** Sep. 2023 - Nov. 2023
Korea Institute of Energy Technology, Naju, South Korea (PI: [Chong-Kwon Kim](#))
Energy AI
- **Backend engineer** Jun. 2018 - Sep. 2018
nTOPAZ, Seoul, South Korea
Back-end Engineer

PUBLICATIONS ([GOOGLE SCHOLAR](#)) - C: CONFERENCE, J: JOURNAL

- (C27) A Differentiable 3D Scene Graph Metric via Contextual Hellinger Triplet Geometry ([link](#))
U-Chae Jun, [Yoonhyuk Choi](#)[†], and Jiwoo Kang[†]
NeurIPS '26
- (J26) Dynamic Sequential Recommendation via Lag-Aware Graph Learning ([link](#))
[Yoonhyuk Choi](#), Jiho Choi, Hawon Shin, Yumin Lee, Chanran Kim, and Chong-Kwon Kim[†]
Information Sciences '26

- (C25) SWORD: Scene-Grounded Wasserstein Object-Relation Distance for 3D Scene Graph Prediction ([link](#))
U-Chae Jun, Jaeun Ko, Yoonhyuk Choi[†], Jiwoo Kang[†]
CIKM '26
- (C24) From Oracle Gains to Repair Priorities: Conservative Split-Conditioned Diagnosis for Modular RAG ([link](#))
Chaewon Sung, Chong-Kwon Kim[†], Yoonhyuk Choi[†]
CIKM '26 (short)
- (C23) Gauge-Equivariant Graph Networks via Self-Interference Cancellation ([link](#))
Yoonhyuk Choi, Jiho Choi, Jiwoo Kang[†]
ICML '26
- (J22) Uncertainty-Aware Adaptive Edge Blocking in Heterophilic Graph Neural Networks ([link](#))
Jiho Choi, Yumin Lee, Chanran Kim, Hawon Shin, Chong-Kwon Kim, Yoonhyuk Choi[†]
ACM TIST '26 (IF: 10.7)
- (J21) Edge-conditioned Markov Kernels for Stable Label Refinement on Heterophilous Graphs ([link](#))
Taewook Ko, Yoonhyuk Choi[†]
Information Sciences '26 (IF: 8.2)
- (C20) Delay-Aware Sequential Recommendation with Dynamic Graphs and Variate-Temporal Decomposition ([link](#))
Yoonhyuk Choi
SIGIR '26 (short)
- (C19) Sheaf Graph Neural Networks via PAC-Bayes Spectral Optimization ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Jongwook Kim, Chong-Kwon Kim[†]
AAAI '26
- (J18) Identifying Heterophilic Neighbors via Confidence-based Subgraph Matching for Graph Neural Networks ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim[†]
Artificial Intelligence '26 (IF: 4.6)
- (J17) LLM-Guided Review Condensation for Text-Convolutional Recommendation ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim[†]
IEEE Access '26 (IF: 3.6)
- (J16) Beyond Binary: Improving Signed Message Passing in Graph Neural Networks for Multi-Class Graphs ([link](#))
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim[†]
IEEE TPAMI '25 (IF: 20.8)
- (J15) Hierarchical Hyperbolic Embeddings for Review-Driven Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim[†]
IEEE Access '25 (IF: 3.6)
- (J14) Generalization of Knowledge Transfer with User Reviews for Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Chong-Kwon Kim[†]
IEEE Access '25 (IF: 3.6)
- (C13) Selective Blocking for Message-Passing Neural Networks on Heterophilic Graphs ([link](#))
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Chong-Kwon Kim[†]
UAI '25
- (C12) Review-Based Hyperbolic Cross-Domain Recommendation ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim[†]
WSDM '25
- (C11) Mitigating Overfitting in Graph Neural Networks via Feature and Hyperplane Perturbation ([link](#))
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim[†]
WSDM '25

- (J10) Beyond Message-Passing: Generalization of Graph Neural Networks via Feature Perturbation for ... [\(link\)](#)
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Chong-Kwon Kim[†]
IEEE TNNLS '24 (IF: 14.25)
- (C9) Improving the Text Convolution Mechanism with Large Language Model for ... [\(link\)](#)
Yoonhyuk Choi, Fahim Tasneema Azad
IEEE Big Data '24 (short)
- (C8) Prioritizing Potential Wetland Areas via Region-to-Region Knowledge Transfer and Adaptive Propagation [\(link\)](#)
Yoonhyuk Choi, Reepal Shah, John Sabo, Selcuk Candan[†], Huan Liu[†]
IEEE Big Data '24
- (C7) Introducing CausalBench: A Flexible Benchmark Framework for Causal Analysis and Machine Learning [\(link\)](#)
Ahmet Kapkıcı, Pratanu Mandal, Shu Wan, Paras Sheth, Abhinav Gorantla, Yoonhyuk Choi, Huan Liu[†], K Selçuk Candan[†]
CIKM '24 (benchmark)
- (C6) Universal Graph Contrastive Learning with a Novel Laplacian Perturbation [\(link\)](#)
Taewook Ko, Yoonhyuk Choi, Chong-Kwon Kim[†]
UAI '23
- (J5) A spectral graph convolution for signed directed graphs via magnetic laplacian [\(link\)](#)
Taewook Ko, Yoonhyuk Choi, Chong-Kwon Kim[†]
Neural Networks '23 (IF: 7.8)
- (J4) Aspect-oriented unsupervised social link inference on user trajectory data [\(link\)](#)
Hyungho Byun, Yoonhyuk Choi, Chong-Kwon Kim[†]
Information Sciences '23 (IF: 8.2)
- (C3) Review-Based Domain Disentanglement without Duplicate Users or Contexts for ... [\(link\)](#)
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim[†]
CIKM '22
- (C2) Finding Heterophilic Neighbors via Confidence-based Subgraph Matching for ... [\(link\)](#)
Yoonhyuk Choi, Jiho Choi, Taewook Ko, Hyungho Byun, Chong-Kwon Kim[†]
CIKM '22
- (J1) Dynamic graph convolutional networks with attention mechanism for rumor detection on social media [\(link\)](#)
Jiho Choi, Taewook Ko, Yoonhyuk Choi, Hyungho Byun, Chong-Kwon Kim[†]
PLOS ONE '21 (IF: 2.9)
.....
- (arXiv) Shifted Graph-Coupled Activation with Learned Edge Targets for Graph Neural Networks
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Jong-Wook Kim, Jiwoo Kang, Chong-Kwon Kim
Under Review (NeurIPS '26)
- (arXiv) Sparse Bayesian Message Passing under Structural Uncertainty
Yoonhyuk Choi, Taewook Ko, Jiho Choi, Jiwoo Kang
Under Review (NeurIPS '26)
- (arXiv) Bias-aware Adaptive Loss for Sequential Recommendation
Yoonhyuk Choi, Jiho Choi, Chong-Kwon Kim
Under Review (CIKM '26)
- (arXiv) Adaptive Branch Specialization in Spectral-Spatial Graph Neural Networks for Certified Robustness
Yoonhyuk Choi, Chong-Kwon Kim
Under Review (TPAMI)
- (arXiv) Learning Sheaf Graph Neural Networks with PAC-Bayesian Spectral Guarantees
Yoonhyuk Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
Under Review (TPAMI)

- (arXiv) Hierarchical and Uncertainty-Aware Graph Neural Networks for Heterophily and Robustness
Yoonhyuk Choi, Chanran Kim, Yumin Lee, Chong-Kwon Kim
Under Review (Pattern Recognition)
- (arXiv) Spectral-Radius and Uncertainty-Aware Blocking in Message-Passing Neural Networks
Yoonhyuk Choi, Jiho Choi, Yumin Lee, Chanran Kim, Chong-Kwon Kim
Under Review (TIST)

INVITED TALKS

- **Co-Week Academy** Jun. 2026
From conventional AI to LLM (PyeongChang, South Korea)
- **DGIST Startup Camp** Nov. 2025
Connecting the dots (Seoul, South Korea)
- **N-EWN Partner Symposium** May. 2024
Identifying Potential Sites for Wetlands (St. Augustine, Florida)

RESEARCH PROJECTS

- **Training Co-Evolving Human-AI to Lead On-Premise Generative Model in Financial Domain** 2026-2029
IITP Korea
– Orchestration, Distillation, and Graph RAG
- **Development of Reliable, Personalized Clinical Decision-Support Technology** 2026-2029
NRF Korea
– LLM and Graph RAG
- **Enhancing Large Language Model with RAG** 2025
Research project, Samsung SDS
– Personalized RAG and knowledge graph construction on Knox Systems
– Knowledge graph construction and RAG for chunk retrieval
– Tiny-LLM (e.g., Llama 3B) distillation with huge-LLM (e.g., Llama 70B)
- **Can Large Language Model Improve the Text Convolution for Review-Based Recommendation?** 2024
Research project, Emit Lab
– Integrated the large language model (e.g., Llama 2, GPT-4) with the text convolution algorithm
– Investigated whether applying summarization based on large language models first, instead of performing 2D convolution on the entire text, results in performance improvement
– Used online shopping mall datasets like Amazon and Walmart
- **Selection Criteria and Assigned Weightage for Identifying Potential Locations Wetland** 2024
Research project funded by NSF (in collaboration with Tulane University)
– Suggested knowledge transfer between different regions and adaptive propagation between grids
– Demonstrated the effectiveness of the framework through real-world scenario
– Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets
- **Causal Discovery of Agricultural Mgmt and Reservoir Op. Induced Water Quality Change** 2023
Research project funded by NSF (in collaboration with University of Arkansas)
– Developed causal discovery algorithm for water quality improvement and reservoir management
– Considered spatial and temporal variations and validated the causal learning ability
– Used Natural Land Cover Dataset (NLCD), Soil Survey Geographic Database (SSURGO) datasets
- **Tracking footprints with graph neural networks for the reduction of virus spread** 2021 - 2022
Coursework project, R&D in AI industry

- Suggested spatial-temporal analysis for the next POI prediction to reduce virus spread
 - Selected as social contributing project
 - Used datasets are Coronamap of South Korea, Gowalla for POI prediction
- Personalized recommendation based on the user’s purchasing histories and social network** 2020
Industry project funded by Samsung Research
- Introduced time series analysis of users’ purchasing history for personalized advertising
 - Applied graph neural networks with binary recommendation techniques
 - Used customer datasets provided by Samsung Research
- Next POI prediction based on user movements collected through large-scale sensors** 2019
Research project funded by Samsung Electronics
- Recommending the next place based on where students visited within Seoul National University
 - Developed energy-saving and effective multi-hop transmission technologies for sensor
 - Collected datasets by attaching special stickers to participants

EXTRACURRICULAR ACTIVITIES

- PC Member** 2026
 NeurIPS / ICLR / IJCAI / ICML / AAAI / KDD (Aug. track) / WSDM / TheWebConf / TPAMI
- PC Member** 2025
 ICLR / IJCAI / ICML / KDD (Feb. track) / CIKM / WSDM / TheWebConf
- PC Member** 2024
 LoG / CIKM / MM
- Reviewer** 2023
 Journal of IEEE Multimedia / Plos one
- Research Assistant (RA), Graduate** Mar. 2019 - Jun. 2021
 Funded by Samsung Research
- Teaching Assistant (TA)** Mar. 2020 - Jun. 2020
 Topic: Social Network Analysis and Anomaly Detection (Advisor: Chong-Kwon Kim)
- Research Assistant (RA), Undergrad** Jun. 2017 - Sep. 2017
 Distributed Computing Lab (Supervisor: Jin-Suk Kim)

AWARDS & GRANTS

- Best Ph.D. Dissertation Award** 2023
 Seoul National University
- Overseas Short-term Training Scholarship** 2023
 Chonnam National University
- BK21 Colloquium Graduate Student Fellowship** 2023
 Seoul National University
- BK21 Star Student Researcher Fellowship** 2023
 Seoul National University
- SIGIR Travel Awards** 2022
 For ACM Student Authors with Accepted Long Paper

•**BK21 Scholarship**
(Graduate) Seoul National University

2020

•**Merit-based Scholarship**
(Undergrad) University of Seoul

2018 - 2019

SKILLS

•**Languages:** Python, C, HTML/CSS

•**Tools / Frameworks:** PyTorch, torch-geometric, Scikit-learn, Git, Django, AWS, LaTeX

REFERENCES

•**ChongKwon Kim:** ckim@kitech.ac.kr

•**TaeKyung Kwon:** tkkwon98@gmail.com

•**U Kang:** ukang@snu.ac.kr